Check for updates

IET Generation, Transmission & Distribution

Research Article

IET Journals
The Institution of Engineering and Technology

DGALab: an extensible software implementation for DGA

ISSN 1751-8687
Received on 5th January 2018
Revised 17th May 2018
Accepted on 19th July 2018
E-First on 4th September 2018
doi: 10.1049/iet-gtd.2018.5564
www.ietdl.org

Saleh I. Ibrahim[1,2], *Sherif S.M. Ghoneim*[2,3] ✉, *Ibrahim B.M. Taha*[2,4]

[1]*Computer Engineering Dept., Faculty of Engineering, Cairo University, Giza, Egypt*
[2]*Electrical Engineering Dept., College of Engineering, Taif University, Taif, KSA*
[3]*Electrical Dept., Faculty of Industrial Education, Suez University, Suez, Egypt*
[4]*Electrical Power and Machines Dept., Faculty of Engineering, Tanta University, Egypt*
✉ *E-mail: sherif.ghoneam@suezuniv.edu.eg*

Abstract: The development of a new dissolved gas analysis (DGA) method often requires a comparative study to assess the accuracy of the proposed technique. This is faced with the following challenges: (i) the time and effort required to implement and validate the implementation of existing DGA methods, adds to the comparative study cost; (ii) the output states of different DGA methods are not similar, which makes it difficult to put methods side by side in a comparative study; and (iii) the availability of test data is limited. In this study, a user-friendly graphical user interface software package, DGALab, is developed to overcome these challenges. DGALab implements a unified DGA diagnosis framework to map the output states of DGA methods to uniform specifications. DGALab includes a library implementing most common DGA techniques, and includes a repository for input datasets available in the literature and collected directly from laboratories. DGALab simplifies the addition of new DGA techniques written in virtually any programming language. As a result, the process of developing a new DGA technique is greatly simplified using DGALab. To evaluate the software package results, the datasets and methods implemented therein were used to regenerate the results published in earlier research papers.

1 Introduction

The dissolved gas analysis (DGA) approach is based on the concentration of the dissolved gases in the insulating oil and the specific limits of dissolved gas ratios. In [1, 2], some gas ratio limits and rules were developed to specify the transformer faults. These rules and gas ratios are assumed based on five main combustible gases: hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), and acetylene (C_2H_2). In some interpretation techniques, the carbon monoxide (CO) was added to the earlier gases. Some un-combustible gases such as carbon dioxide (CO_2), nitrogen (N_2), and oxygen (O_2) were developed during faults but they do not affect the fault-type diagnosis.

Different approaches of the DGA technique were reported in the standards such as Dornenburg method, Rogers' three and four ratios methods, and IEC standard code [1, 2]. These methods experienced some problems to identify the transformer faults and suffered in some cases from no-decision status. Some graphical methods were also proposed based on the gases dissolved in the transformer oil. The common graphical techniques were attributed to the Duval triangle [3, 4] and the pentagon [5]. Duval triangle and pentagon were used after checking the status of the oil by another DGA method to ensure the existence of a fault condition. Duval triangle was based on the concentration percentage according to three gases (CH_4, C_2H_4, and C_2H_2) but the pentagon technique was based on five gases (H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2).

Recently, a number of artificial intelligence techniques had been applied with DGA, in order to develop new techniques to interpret transformer faults. These techniques were based on artificial neural networks (ANNs) [6–8], fuzzy logic [9–11], support vector machine [12, 13], as well as new approaches that were contributed to solve the no-decision problems [14–17].

There were a few commercial software packages available to perform DGA. 'Inside View' [18], according to the company's website, supports the following DGA methods: Rogers Ratios, IEC 60599 Ratios, Dornenburg Ratios, Key Gas Method, IEEE Recommendations, Trends and Detailed Duval Triangle Analysis. 'Serveron' [19] included IEEE and IEC diagnostic tools such as Rogers Ratios, and the Duval's Triangles and Duval's Pentagon for advanced diagnostics. DGA Expert Systems Software version 3 [20] by Northern Technology & Testing took the ppm data that generated during the transformer faults and interpreted the transformer faults. It used an Excel file based on three interpretive methods such as Key gas, Rogers' ratios, and IEEE/ANSI C57.104 Guideline. It can be used in conjunction with chromatography software via external data import. DGA Viewer software [21] can interpret the online DGA results. It made condition-based maintenance by using DGA results in several platforms. It used the Duval triangle rules to interpret the transformer faults. GE Perception Fleet [22] is smart software that helps to transform fleet management assessment. It evaluates and assesses the risk of the transformer failure and it can be used via Internet. Perception Fleet's standards-based algorithm includes: CIGRE TB 227 which provides five transformer conditions in terms of its reliable operation, IEEE C57.104 which determine the condition level of the transformer based on the total dissolved combustion gases (TDCG) and the variation of the gases rate per day and provide the time of the next test, IEC 60599 which provides the rate of change of the dissolved gases for normally operating transformer and finally IEC 60422 that used to analyse the moisture on the oil. However, commercial software packages did not target the research community and, therefore, did not support user-defined methods and were not designed to simplify comparative accuracy analysis or provide a platform to build a corpus of datasets for testing new techniques. In addition, it did not provide an updated DGA technique that used artificial intelligence to enhance the diagnosis accuracy of the tradition DGA techniques.

In the research area, several publications discussed the software packages to interpret the transformer faults but all of them were based on traditional DGA techniques. Singh *et al.* [23] provided Duval triangle methods' software which was implemented using MATLABs to diagnose and interpret the transformer faults. Victor and Marungsri [24] presented a computer-based software with a database for transformer condition monitoring with DGA suing

© The Institution of Engineering and Technology 2018

five DGA techniques such as (Key Gas method, Dornenburg ratio method, Roger's ratio method, IEC ratio method, and Duval triangle method). Suleiman *et al.* [25] provided a computer programme called Hybrid-DGA interpretation software that was able to enhance the accuracy of DGA techniques based on a combined common DGA techniques such as Key Gas, Rogers ratio, IEC ratio, and Duval triangle. On the basis of [23–25], all of them constructed their software based on traditional DGA techniques which provided low accuracy for transformer fault diagnosis, it did not present any of new techniques which combine the artificial intelligence with the traditional DGA techniques to improve the diagnosis accuracy. Moreover, their programme cannot help the other to benefit from the comparative purpose between newly proposed techniques with their work to save the effort and time for the other researchers.

The main contribution of this paper is to provide the first extensible DGA software package (DGALab) intended to reduce researchers' programming effort. DGALab provides a user-friendly tool to implement DGA comparative accuracy analysis studies. It is a framework which contains several DGA techniques; traditional DGA techniques such as Duval triangle method [3, 4], Rogers' 4-ratios method [1, 2], IEC 60599 method [1], Refining IEC method [26], and refining Rogers' method [20], in addition to the traditional DGA methods combined with the artificial intelligent techniques to enhance the accuracy of the traditional DGA methods [6, 7, 14, 15]. It was constructed to help the researchers to compare the accuracy of their new proposed DGA techniques with the existence and published DGA techniques. Therefore, DGALab software can save their effort and time for the comparison process. Measuring the accuracy of each DGA technique is a very important feature in DGALab to save a comparison environment for any new proposed DGA technique.

The main features of the DGALab software package are as follows:

i A user-friendly graphical user interface (GUI) for specifying input gas ratio data and selecting which DGA methods to apply.
ii Built-in implementation of the following DGA methods: Duval, Rogers' 4-ratio, IEC 60599 ratios, refined Rogers' 4-ratio, refined IEC 60599, clustering, ANN, and conditional probability [6, 14, 15], all of these methods were demonstrated in the Appendix.
iii The ability to integrate with other DGA method written in any programming language.
iv An extensible database of datasets populated by datasets made available by earlier DGA research literature.
v Results of accuracy analysis in both tabular and graphical formats.
vi Real-time application to diagnose transformer fault type using DGA methods included in DGALab, through the single point input mode.

In the DGALab, two common DGA methods were used (IEC standard [1], Duval triangle method [3, 4]), and the rest methods in software package (DGALab) were based on the confirmed published DGA technique as clustering in [14], conditional probability technique in [15], California State University Sacramento (CSUS) ANN technique in [6, 7], and some of modifications of IEC standard and Rogers' techniques as in [20]. The accuracy of each of the previous DGA technique was reported in [6, 14, 15, 26] by using 386 samples that were collected from laboratory results and the literature.

In DGALab, the source of each dataset was identified to facilitate the referencing process. The reference for each dataset was mentioned as declared in Section 5. Only 240 samples were collected from the central chemical laboratory which belongs to the Electric Transmission and Distribution Company in Egypt. The rest of the datasets were collected from the literature. All data were categorised and explained in [14].

Although DGALab is focused mainly on producing comparative studies for future DGA techniques, DGALab can also be used for real-time practical diagnosis of transformer faults. The gas concentrations of an oil sample taken from faulted transformer can be fed into DGALab through the 'Single Point' input mode to be analysed through any or all of the available DGA methods to identify the fault type of the transformer. A sample of the transformer oil could be obtained and sent to the central chemical laboratory to analyse the dissolved gases if a malfunction of the transformer is suspected. This may take some time and a breakdown in the transformer insulation may result in the combustion of the transformer and cause severe losses in the electrical grid. Therefore, the real-time application is very important to maintain the transformer and the operation of the power network. In the previous work created in another research to one of the authors, an intelligent model of how to deal with the fault in the transformer, if any, where the concentration of dissolved gases is continuously determined for the transformer oil and diagnosed the fault type and its severity. Then, it gave a signal to the control room in the main substation (secondary station) with the current transformer status and identified the action and the next time to test the transformer [27].

The transformer fault severity categorised to three grads based on the degree of the danger (low, medium, and high fault severity), in addition, the TDCG played a contribution to identify the transformer condition and the action was taken based on these two states: the fault severity and the TDCG. Assume that the rate of change of TDCG is >30 ppm, the fault was a high thermal fault (T3) and the calculated severity is high, then the appropriate action must update the main control centre (primary substation) by the message of new analysis after a day. Hence the intelligent proposed system must inform the secondary substation with a message including the corresponding information which will be evaluated by the primary station and then the action will be accepted to call more information or modify the action. Therefore, the DGALab is very important to identify the fault type and calculate the TDCG and give a message to the intelligent system to calculate the fault severity and transfer a message to the primary station to take an appropriate action which including the decision to trip the transformer or maintain it in the network. If the transformer main in the network, when must the next test be done. Measuring device for DGA can be mounted on the transformer to test an oil sample every certain time and the DGALab can associate to read the test sample results. Furthermore, it provides the required information including the fault type, its severity, and TDCG, then transfers these data to the secondary substation and the primary substation. The action will be decided based on the danger level of the fault and the decision whether the transformer trip or not as well as the next appropriate test of the transformer oil.

In the following, we discuss in detail the design of DGALab. First, a unified diagnosis framework is presented. Then, the choice of development platform, i.e. operating system and development environment, is explored. The user interface design issues are then discussed. Then, the chosen approach to an extensible programming interface is presented. Moreover finally, we present some use cases to illustrate the usefulness and validity of DGALab in various practical and research scenarios.

2 Unified DGA diagnosis framework

To achieve the first design goal, namely to enable the incorporation of the most common DGA methods as a basis for comparison with new methods, a unified diagnosis framework had to be developed. In particular, the inputs and outputs of the DGA diagnosis process had to be uniformly specified. Existing methods, then, have to be adapted to transform their inputs and outputs to this unified framework.

To achieve maximum inclusiveness, the unified framework defines the input as nine optional gas concentration ratios, namely H_2, CH_4, C_2H_6, C_2H_4, C_2H_2, CO, CO_2, N_2, and O_2. Any missing input is represented as a value of (−1). When a DGA method is provided with input in this format, it can verify the availability of all necessary input components and ignore unneeded ones. The output, on the other hand, is defined to allow six fault diagnosis codes.

In both the clustering and the conditional probability methods, the data were categorised to refer to six faults such as, partial discharge (PD), low-energy discharge (D1), high-energy discharge

© The Institution of Engineering and Technology 2018

17518695, 2018, 18, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-gtd.2018.5564 by Capes, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

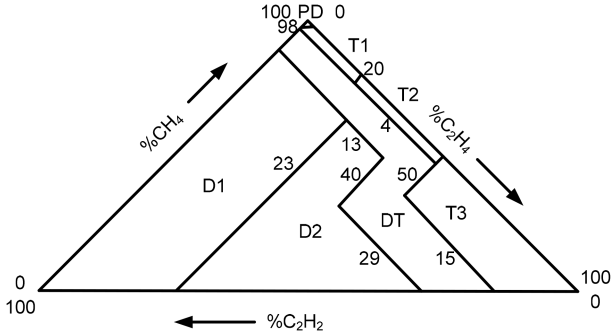


Fig. 1 *Duval triangle as a diagnostic tool to detect incipient faults in transformers*

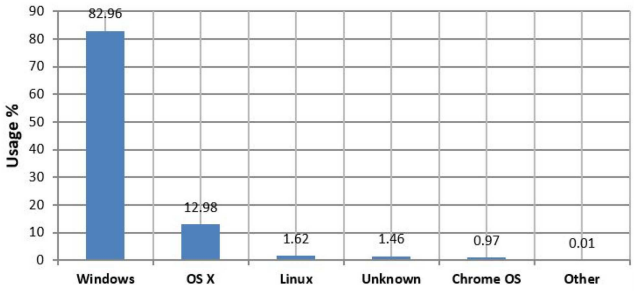


Fig. 2 *Desktop operating system market share according to StatCounter for October 2017*

(D2), low thermal fault (T1), medium thermal fault (T2), and eventually high thermal fault (T3). Table 1 indicates eight fault types including the normal operation and the state of no-fault identification for IEC code. The eight fault types were arranged to explain only the six fault types; the state (s.3) refers to PD, state (s.4) and (s.5) refer to D1, state (s.6) refers to D2, the states (s.7) and (s.8) refer to a low thermal fault (T1), state (s.9) refers to T2, and state (s.10) refers to T3. The fault types and fault codes of Rogers' four ratios method were also illustrated in Table 1. The states (s.4), (s.5), and (s.6) were merged to refer to low thermal fault (T1), the states (s.7) and (s.8) were merged to refer to the medium thermal fault (T2), and the state (s.11) refers to the high-energy discharge fault (D2), state (s.3) refers to the PD fault, states (s.10), s.12), and (s.13) refer to low-energy discharge D1 and state (s.9) refers to high thermal fault T3. In Duval triangle method, the electrical-thermal fault (DT) one is divided between the faults D1 and D2 to be able to make a comparison with the other methods. The boundaries of each zone for the Duval triangle are shown in Fig. 1.

3 Platform and development environment choice

DGALab was developed and tested in MATLAB IDE on Microsoft Windows operating system. Windows continues to lead the desktop operating systems market and is assumed to be the most widely used operating system among researchers in the field. According to a study by StatCounter web traffic analysis tool, shown in Fig. 2, over 80% of desktop operating systems browsing the Internet in October 2017 is equipped with one version or another of Windows.

The choice to implement DGALab in MATLAB is justified by the fact that the MATLAB programming language and the associated library is commonly used in the research community. MATLAB language has gained this particular place in the academia because of its ease of use and its powerful toolboxes and libraries. The success of MATLAB has inspired a number of alternative open sources products such as GNU Octave and Scilab. The research community, however, is gradually embracing more open computing platforms such as Unix-based operating systems and open source products such as Octave and Scilab. Therefore, the portability of DGALab to other operating systems and open source alternative libraries such as GNU Octave was investigated. The porting was successful to a great extent, with a small sacrifice of dropping the 'uitable' element which allows DGALab to display tabular results on screen.

As of this moment, numerical results could only be viewed in tabular form after being exported to an Excel worksheet. It is expected that the 'uitable' functionality will eventually be contributed to GNU Octave, thus allowing DGALab to function fully therein.

Table 1 Fault diagnosis for IEC code and Rogers' four ratios [28]

	Fault type	
proposed fault type	IEC code	Rogers' four ratios
No Fault	normal state (s.1)	normal state (s.1)
Undetermined fault	no-fault identification (s.2)	no-fault identification (s.2)
PD	low-energy PD (s.3)	PD (s.3)
D1	high-energy PD (s.4)	arcing discharge with low energy (s.10)
	low-energy discharge (s.5)	continuous spark (s.12)
		PD with tracking (s.13)
D2	high-energy discharge (s.6)	arc with a high-energy density (s.11)
T1	low thermal fault <150°C (s.7)	thermal fault <150°C (s.4)
	thermal fault (150–300°C) (s.8)	thermal fault (150–200°C) (s.5)
		thermal fault (200–300°C) (s.6)
T2	medium thermal fault (300–700°C) (s.9)	thermal fault (increase in overall temperature in the conductive parts) (s.7)
		thermal fault (circulating current in windings) (s.8)
T3	high thermal fault >700°C (s.10)	thermal fault (core and tank circulating currents, overheated joints) (s.9)

4 GUI design

The software package comes with an intuitive GUI that can be used with little or no need for an instruction manual. The GUI simplifies the user's job of using the software. Tooltips and informative message boxes were added and were needed to guide the user through the process. Through the GUI, the user can choose the DGA methods to be applied and the data point or the datasets to which the methods will be applied. Then, DGALab displays the analysis results both as a bar chart and in tabular format as in the results and discussion section. Moreover, the software package allows the user to import test datasets and to export analysis results through easily accessible graphical controls. Fig. 3 illustrates the flowchart DGALab's main procedures.

The flowchart in Fig. 3 illustrates DGALab's main procedure. Before the main procedure can run, the user must use the GUI to select the DGA methods to be applied, the input mode, whether single point or dataset mode, and the gas concentration data to which DGA methods will be applied. When the user clicks on the 'RUN' button, the procedure illustrated in Fig. 3 is performed. Essentially, each of the selected DGA methods is applied to the input data to produce a corresponding fault diagnosis. If the multipoint dataset mode is selected, the procedure combines the data points found in the datasets chosen through the GUI into one dataset before applying DGA methods, one by one, to the combined input dataset. If the input dataset includes the actual fault diagnosis, the programme performs accuracy analysis and generates the corresponding accuracy analysis tables and graph. Finally, the programme displays the results including fault-type diagnosis and any accuracy analysis figures. The procedure is illustrated in more details later in Section 6.

© The Institution of Engineering and Technology 2018

17518695, 2018, 18, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-gtd.2018.5564 by Capes, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

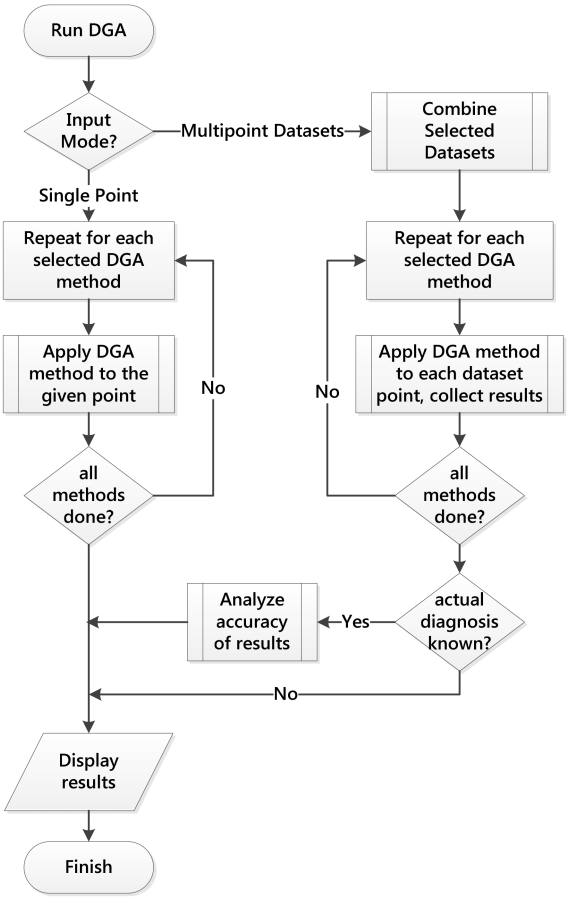


Fig. 3 *Flowchart of DGALab main procedure*

5 Programmable extensibility approach

A user wishing to study a new DGA method needs a way to include the implementation of the new method into the software package. DGALab provides two simple ways of integrating user-defined code into the software package, depending on the programming language of implementation. If the researcher is using MATLAB-compatible software for developing a new method, the new code can be integrated as a MATLAB script with specific input and output (I/O) variables, as shown in Fig. 4*a*.

If the user prefers another programming language, the user-defined code can be integrated as a stand-alone executable file with which the DGA tool communicates through a file I/O stream interface.

Using a simple text-file I/O stream interface guarantees compatibility with all programming languages. I/O files are automatically mapped to console input stream and console output stream, respectively, making the integration of new methods developed in any language quite a straight-forward exercise. Basically, an implementation of a new method is required to read the dissolved gas concentration ratios from the console input stream and to write the diagnosis code into the console output stream as illustrated in Fig. 4*b*.

The input file is prepared automatically by the DGA software package before invoking the user-defined method. In addition, when the user-defined methods are done computing, the software package reads its output from the corresponding file. In both cases, the new method will need to be listed in the DGA software package GUI, and this can be done by adding its name in a configuration file (config.xls) using the 'Configure Methods' functionality from the GUI or 'Cfg_Methods' from the command window.

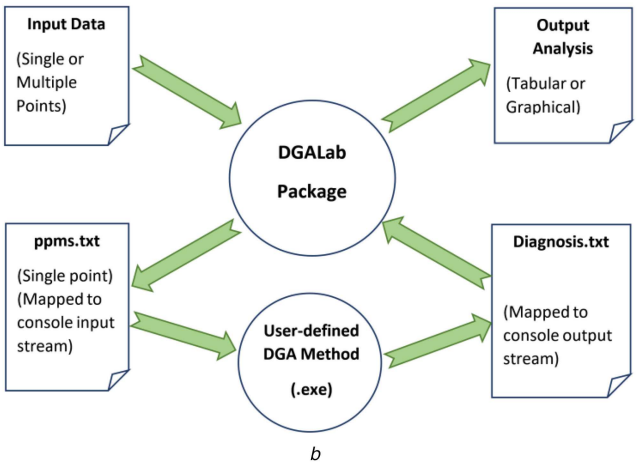
The implementation file name extension of a method determines whether DGALab will treat it as a MATLAB source code or a stand-alone executable. Namely, the filename extension is either '.m' or '.exe', as shown in Fig. 5. For instance, the Rogers' method is implemented in MATLAB, so its

```
%Input:
%------
%The input is the gas concentrations
%stored in variables named: h2, ch4,
%c2h6,  c2h4,  c2h2, co, co2, n2, o2
%and vector:  ppms(1:9) respectively.
%Unused gases are given the value -1

% Analysis
%---------
% Implement your method here

%Output:
%-------
%Set variable Diagnosis to 0,1, ...,7
%based on the diagnosed fault code,
%0=NF,1=PD,2=D1,3=D2,4=T1,5=T2,6=T3,7=UD
```

a



b

Fig. 4 *Integration with a user-defined method implemented*
(a) MATLAB, ***(b)*** Stand-alone executable (.exe)

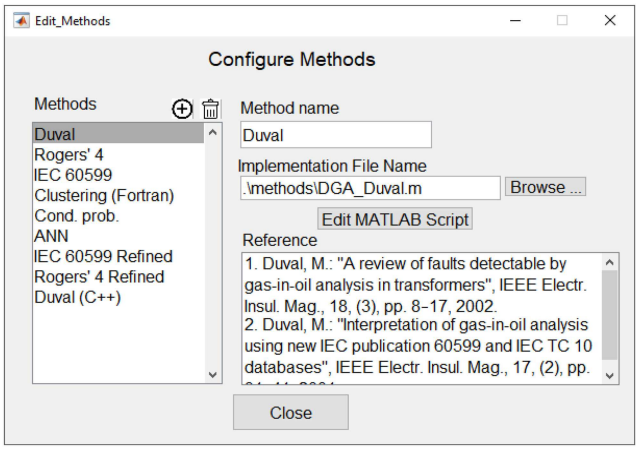


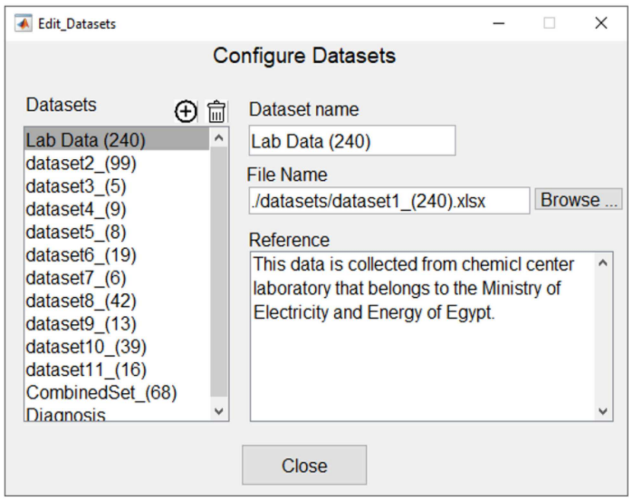
Fig. 5 *Configuring the list of methods integrated into DGALab*

implementation file is in the '.m' format, while the clustering method [14] is implemented in Fortran and runs as a stand-alone executable, so its implementation file is in the '.exe' format. The user is responsible for compiling the source code into a stand-alone executable using the compiler appropriate for the programming language of choice and the target platform.

The DGALab programme benefits can be categorised as evaluating the accuracy of DGA techniques using reference sample with known actual diagnoses (Lab and literature samples) and diagnose the fault type of new sample that the actual diagnosis was unknown. Therefore, the user can configure the list of datasets available in DGALab as shown in Fig. 6*a*. In addition, the user can add a new dataset (*n* number of available data) in the form of an Excel worksheet with a very simple format similar to the existing

© The Institution of Engineering and Technology 2018

17518695, 2018, 18, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-gtd.2018.5564 by Capes, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



a

	A	B	C	D	E	F
1	H2	CH4	C2H6	C2H4	X2H2	ACT
2	130	140	2	120	0.001	4
3	55	22	358	2.6	0.001	4
4	78	66	283	2.6	0.001	4
5	16000	3600	670	14	0.001	1
6	6600	1000	38	2	19	1

b

Fig. 6 *Adding a new dataset and its file format*
(a) Configuring datasets, to add new datasets to the corpus, ***(b)*** Sample dataset file format

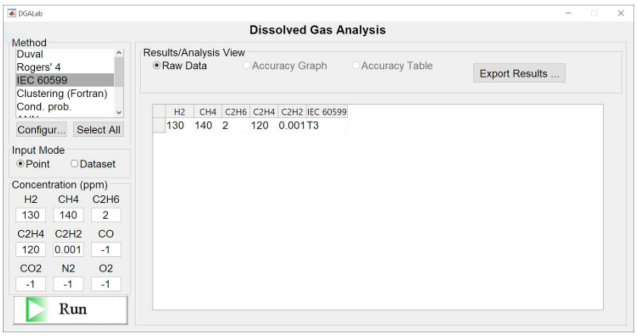


Fig. 7 *Scenario #1: Apply IEC method to a single point and display diagnosis*

datasets, as shown in Fig. 6*b*. The first row of the sheet contains the provided gas names and an actual fault-type 'ACT' column when the accuracy of the DGA method is intended for the comparison purpose with other DGA methods. Gases which are not provided are assumed to be unknown and represented as −1 when provided to the diagnosis method implementation. If the dataset includes an actual diagnosis for the purpose of accuracy analysis, the column titled 'ACT' contains the diagnosis code, if not; the gases concentration for any number of data was recorded and with the selected DGA method the diagnosis for these data was obtained.

6 Distribution

DGALab software package, as well as its accompanying documentation, is available at an online repository hosted on GitHub [29]. DGALab comes with a number of examples showing how to implement a new method in MATLAB, C++, and Fortran languages. Researchers are also encouraged to contribute methods and datasets to the online repository. Contributions demonstrating the use of other programming languages [29] are invited.



a

b

c

Fig. 8 *Scenario #2*
(a) Applying a selected method to a selected dataset, ***(b)*** Accuracy graph, ***(c)*** Accuracy analysis table

7 Results and discussion

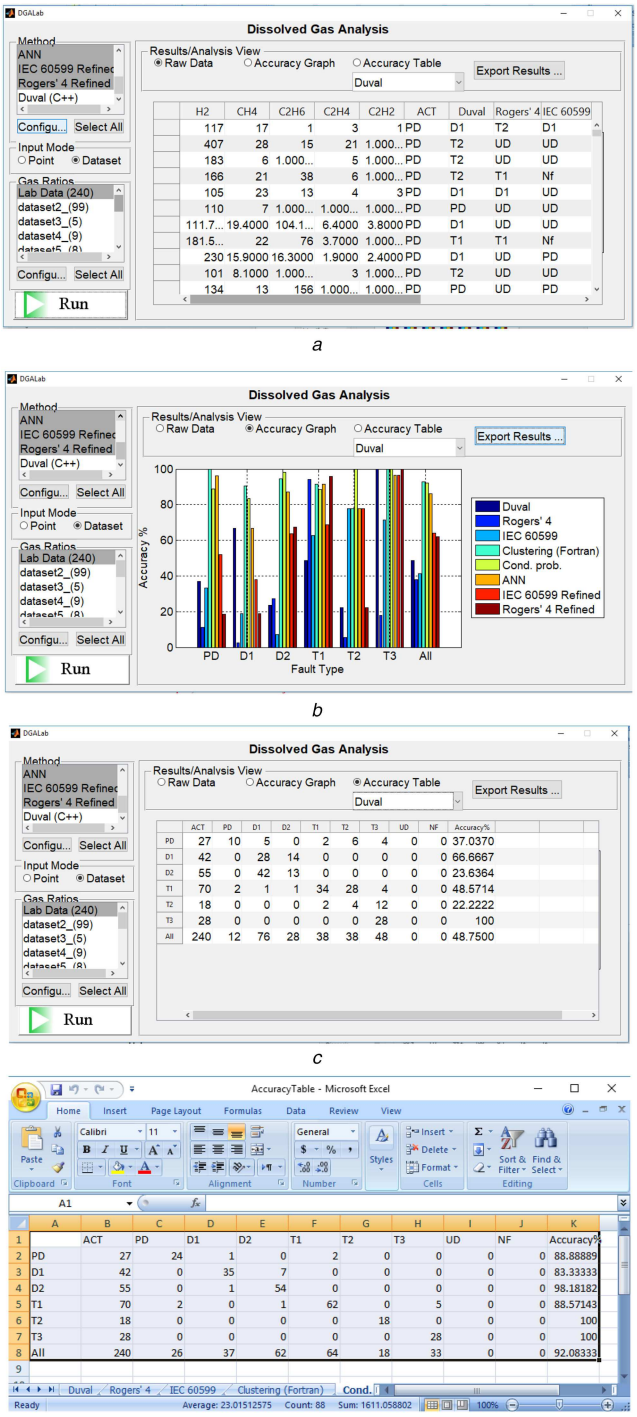
To demonstrate the capabilities of the prepared 'DGALab' software package, three experiments were performed. The first experiment, shown in Fig. 7, illustrates a simple use-case scenario, in which DGALab is used to apply one DGA method to a single data point of gas concentrations and display the corresponding transformer fault diagnosis. The user selected the IEC method from the methods list and supplied the concentration of gases in the bottom-left text boxes after choosing the 'Single Point' option. The concentration of only five gases is supplied, while the remaining four were omitted. Omitted values are supplied as negative values (−1) to indicate they should not be displayed in results. After pressing the 'Run' button, DGALab displays the analysis results of the selected method in a separate column at the right of the 'Raw Data' table.

The second experiment, shown in Fig. 8*a*, illustrates a more elaborate use-case scenario, in which DGALab is used to apply one DGA method to a selected dataset of gas concentrations. Since the dataset includes an actual diagnosis, an accuracy analysis can be performed and displayed both in tabular view, Fig. 8*b*, and the graphical view, Fig. 8*c*.

The accuracy of each DGA method was computed based on the agreement between the results from the suggested DGA method and the actual faults whether for the samples that were collected from the laboratory or the samples were collected from the literature. If the DGA technique result of each sample agrees with the actual fault of the sample, then the accuracy is considered 100% and if not the accuracy is computed as null. As seen in

© The Institution of Engineering and Technology 2018

17518695, 2018, 18, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-gtd.2018.5564 by Capes, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



a

b

c

d

Fig. 9 *Scenario #3*

(a) Apply selected methods to selected datasets, ***(b)*** Accuracy analysis graph, ***(c)*** Accuracy analysis table for the method selected, ***(d)*** Excel file containing the accuracy of each DGA method according to each fault type

Figs. 8*a*–*c*, the laboratory dataset (240 samples) was applied with the ANN technique (CSUS method). In Fig. 8*a*, the results of CSUS with ANN technique were shown and then the results were compared with the actual faults and then Figs. 8*b* and *c* showed the accuracy of the CSUS ANN technique for each fault. For example, the accuracy to detect PD with the suggested DGA technique (CSUS ANN) was 96.29% and for T2 was 77.77%. In addition, the accuracy of the suggested DGA technique (CSUS ANN) for all detected data was 86.26%.

The third experiment demonstrates a comparative accuracy study. A selection of DGA methods is applied to a selected set of gas concentration datasets as a 240 laboratory sample, the diagnosis produced by each method is shown in the 'Raw Data' table in Fig. 9*a*. Fig. 9*a* illustrates that all DGA techniques were applied to a 240 samples data and the diagnosis results of each DGA technique for each sample in 240 samples data were determined as in the column Duval, Rogers' 4, IEC 60599, and the other techniques which do not appear in table because of the small width of the shown table. In MATLAB, using scroll button, everything can be shown whether the selected DGA techniques or the diagnosis results of each technique.

A comparison of the accuracy of the applied methods can be displayed as a bar chart, which can optionally be exported to an image file, as shown in Fig. 9*b*. Fig. 9*b* illustrates the accuracy of each DGA techniques according to the fault types. The DGALab explored two types of accuracies, the first one is the accuracy of fault type for certain DGA technique and the other is the overall accuracy of the certain DGA technique including all fault types. The two accuracy types can be calculated as follows: (i) the fault-type accuracy for each DGA technique = number of succeeded diagnosis for this fault-type/total number of fault-type samples and (ii) the overall accuracy of the DGA technique = number of succeeded diagnosis for all fault types/total number of applied samples.

According to Fig. 9*c*, the Duval triangle succeeded to diagnose the fault types as follows; PD (10/27), D1(28/42), D2(13/55), T1(34/70), T2(4/18), and T3(28/28). Therefore, the accuracy for each fault-type diagnosis is 37.037% for PD, 66.66% for D1, 23.68% for D2, 38.57% for T1, 22.22% for T2, and 100% for T3. In addition, the overall accuracy to diagnose the corrected fault is 48.75% (117/240).

It was also cleared from Fig. 9*b* that the accuracies of different methods for detecting PD fault were 33.33% for IEC, 100% for clustering, 88.88% for conditional probability method, 37.03% for Duval triangle, 11.11% for Rogers' 4-ratios, 96.29% for ANN technique, 51.85% for refining IEC, and 18.5% for refining Rogers' 4-ratios.

The accuracy analysis can be displayed in table form for each method as shown in Fig. 9*c*. By changing the method selected in the dropdown list at the top-right corner, the accuracy table for the selected method is displayed.

In Fig. 9*c*, for instance, the 'Duval triangle' method is selected, so the shown table reflects the accuracy analysis table for that method. The accuracy of each method with the selected 240 samples can be saved in an Excel file which includes several sheets containing the accuracy for each DGA method with each fault type as shown in Fig. 9*d*.

To demonstrate the validity of the DGALab software package, the programme was applied to test data collected from published articles. The results produced by DGALab were then compared with published results. Fig. 10*a* illustrates the data that were collected from [14] and its original results. Fig. 10*b* shows the output of DGALab when applying the same method, which was 'Clustering (Fortran)', to the same data.

The datasets obtained from [14] was added to DGALab and selected for the experiment shown in Fig. 10*b*. The concentration of each dissolved gases appears in the first five columns and the actual faults appear in the actual (ACT) column.

In addition, the results of the software package appear under the 'Clustering (Fortran)' column. The diagnosis results produced by DGALab, shown in Fig. 10*b*, match the original results that were shown in Fig. 10*a*. The diagnosis accuracy for each fault of the input data can be obtained in the graph and tabulated form.

At the rightmost side, the overall accuracy percentage is listed. For example, the number of samples that refer to the actual PD fault is two and the results from the next eight columns refer that the number of samples that were correctly detected was two. On the other hand, the number of samples that the actual faults are high-energy discharge is two and the next eight columns show that one sample is correctly identified as high-energy discharge and the other one was incorrectly diagnosed as low-energy discharge.

8 Conclusion and future work

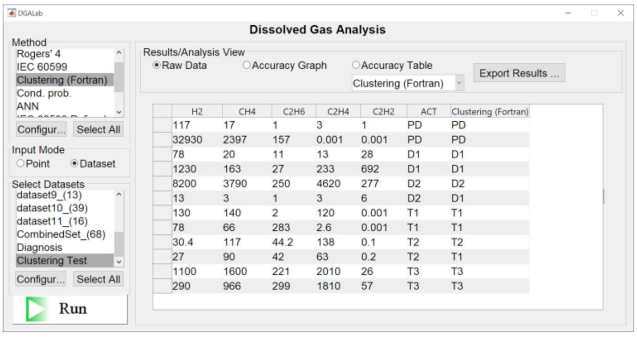
DGALab is a DGA software package with demonstrated usability for performing DGA. DGALab is considered the first DGA

© The Institution of Engineering and Technology 2018

17518695, 2018, 18, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-gtd.2018.5564 by Capes, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Gases concentration (ppm)					Actual	Modified new approach DGA
H_2	CH_4	C_2H_6	C_2H_4	C_2H_2		
117	17	1	3	1	PD	PD
32930	2397	157	0.001	0.001	PD	PD
78	20	11	13	28	D1	D1
1230	163	27	233	692	D1	D1
8200	3790	250	4620	277	D2	D2
13	3	1	3	6	D2	D1
130	140	2	120	0.001	T1	T1
78	66	283	2.6	0.001	T1	T1
30.4	117	44.2	138	0.1	T2	T2
27	90	42	63	0.2	T2	T1
1100	1600	221	2010	26	T3	T3
290	966	299	1810	57	T3	T3

a



b

Fig. 10 *Comparing data and results from literature with the results of the DGALab*

(a) Sample data in [14] and its results, ***(b)*** The same data results using DGALab

software package for academic research with unique extensibility features. It is important to note that the goal of this work was not to evaluate any particular DGA method but it was rather to provide the DGA research community with an easy to use platform through which they can perform comparative studies to evaluate 'their own' novel methods against existing methods included in DGALab package or any other methods added to the package later on. The experiments performed using DGALab and reported in this paper were intended to illustrate the ease of use and the quality of the tables and graphs produced, rather than to draw any conclusions regarding the accuracy of any particular DGA method. The results presented in this paper show how DGALab satisfies its major design goals, namely functionality, usability, and extensibility. DGALab has also been validated by regenerating results already published in DGA literature and showing that DGALab's results match previous results.

DGALab has the potential to be the standardised platform for developing DGA-based transformer fault diagnosis methods and evaluation datasets. This is why future work should focus on completing the set of integrated DGA diagnosis methods and growing the testing data corpus. Future work should also include porting and testing DGALab to open source operating systems and development environments to maximise DGALab's availability to the research community.

9 References

[1] IEC 60599: '*Mineral oil-filled electrical equipment in service - Guidance on the interpretation of dissolved and free gases analysis*', (IEC, Geneva, Switzerland), Edition 2.1, 2007-05

[2] IEEE Guide for the Interpretation of Gases Generated in Oil-Immersed Transformers, IEEE Standard C57.104-2008, February 2009

[3] Duval, M.: 'A review of faults detectable by gas-in-oil analysis in transformers', *IEEE Electr. Insul. Mag.*, 2002, **18**, (3), pp. 8–17

[4] Duval, M.: 'Interpretation of gas-in-oil analysis using new IEC publication 60599 and IEC TC 10 databases', *IEEE Electr. Insul. Mag.*, 2001, **17**, (2), pp. 31–41

[5] Mansour, D.A.: 'Development of a new graphical technique for dissolved gas analysis in power transformers based on the five combustible gases', *IEEE Trans. Dielectr. Electr. Insul.*, 2015, **22**, (5), pp. 2507–2512

[6] Ghoneim, S.S.M., Taha, I.B.M., Elkalashy, N.I.: 'Integrated ANN-based proactive fault diagnostic scheme for power transformers using dissolved gas analysis', *IEEE Trans. Dielectr. Electr. Insul.*, 2016, **23**, (3), pp. 1838–1845

[7] Guardado, J.L., Nared, J.L., Moreno, P., *et al.*: 'A comparative study of neural network efficiency in power transformers diagnosis using dissolved gas analysis', *IEEE Trans. Power Deliv.*, 2001, **16**, (4), pp. 643–647

[8] Miranda, V., Garez Castro, A.R., Lima, S.: 'Diagnosing faults in power transformers with autoassociative neural networks and mean shift', *IEEE Trans. Power Deliv.*, 2012, **27**, (3), pp. 1350–1357

[9] Huang, Y.-C., Sun, H.-C.: 'Dissolved gas analysis of mineral oil for power transformer fault diagnosis using fuzzy logic', *IEEE Trans. Dielectr. Electr. Insul.*, 2013, **20**, (3), pp. 974–981

[10] Hooshmand, R., Banejad, M.: 'Application of fuzzy logic in fault diagnosis in transformers using dissolved gas based on different standards', *World Acad. Sci. Eng. Technol.*, 2006, **17**, pp. 157–161

[11] Taha, I.B.M., Ghoneim, S.S.M., Zaini, H.G.: 'A fuzzy diagnostic system for incipient transformer faults based on DGA of the insulating transformer oils', *Int. Rev. Electr. Eng. (I.R.E.E.)*, 2016, **11**, (3), pp. 305–313

[12] Bacha, K., Souahlia, S., Gossa, M.: 'Power transformer fault diagnosis based on dissolved gas analysis by support vector machine', *Electr. Power Syst. Res.*, 2012, **83**, (1), pp. 73–79

[13] Wei, C., Tang, W., Wu, Q.: 'Dissolved gas analysis method based on novel feature prioritisation and support vector machine', *IET Electr. Power Appl.*, 2014, **8**, (8), pp. 320–328

[14] Ghoneim, S.S.M., Taha, I.B.M.: 'A new approach of DGA interpretation technique for transformer fault diagnosis', *Int. J. Electr. Power Energy Syst.*, 2016, **81**, pp. 265–274

[15] Taha, I.B.M., Mansour, D.A., Ghoneim, S.S.M., *et al.*: 'Conditional probability-based interpretation of dissolved gas analysis for transformer incipient faults', *IET Gener. Transm. Distrib.*, 2016, **11**, (4), pp. 943–951

[16] Morais, D.R., Rolim, J.G.: 'A hybrid tool for detection of incipient faults in transformers based on the dissolved gas analysis of insulating oil', *IEEE Trans. Power Deliv.*, 2006, **21**, (2), pp. 673–680

[17] Abu-Siada, A., Islam, S.: 'A New approach to identify power transformer criticality and asset management decision based on dissolved gas-in-oil analysis', *IEEE Trans. Dielectr. Electr. Insul.*, 2012, **19**, (3), pp. 1007–1012

[18] M. Schaffer 'Inside View' commercial software package. Available at https://www.morganschaffer.com/page-inside_view.html, accessed March 2017

[19] Qualitrol 'Serveron' commercial software package. Available at http://www.qualitrolcorp.com/products/dissolved-gas-analyzers/multi-gas-analyzers/serveron-tm8-multi-gas-on-line-dissolved-gas-monitor/, accessed March 2017

[20] DGA Expert Systems Software version 3 By Northern Technology & Testing. Available at http://www.nttworldwide.com/dgasoftware.htm, accessed May 2018

[21] Smart DGA Monitoring Solutions. Available at https://www.lumasenseinc.com/uploads/Products/Gas_Sensing/SmartDGA_for_Transformers/SmartDGA_Products/pdf/EN-SmartDGA-Monitoring-Solutions_Brochure.pdf, accessed May 2018

[22] GE Perception Fleet. Available at http://www.gae.co.id/detail/ge-perception-fleet-284, accessed May 2018

[23] Singh, S., Joshi, D., Bandyopadhyay, M.N.: 'Software implementation of Duval triangle technique for DGA in power transformers', *Int. J. Electr. Eng.*, 2011, **4**, (5), pp. 529–540

[24] Victor, P., Marungsri, B.: 'Implementation of computer based software for oil immersed power transformer condition monitoring via dissolved gas-in-oil results'. IEEE PES Thailand Joint Symp. Advanced Technology in Power Systems, Chulalongkorn, Indonesia, 11 March 2016, pp. 29–35

[25] Suleiman, A.A., Muhamad, N.A., Bashir, N., *et al.*: 'Introducing the hybrid-DGA interpretation software as an effective power transformer management tool'. Fourth Int. Conf. Power Engineering, Energy and Electrical Drives, Istanbul, Turkey, 13–17 May 2013, pp. 1410–1414

[26] Taha, I.B.M., Ghoneim, S.M., Duaywah, A.S.A.: 'Refining DGA methods of IEC code and Rogers four ratios for transformer fault diagnosis'. 2016 IEEE PES General Meeting, Boston, USA, 17–21 July 2016

[27] Ghoneim, S.S.M.: 'Intelligent prediction of transformer faults and severities based on dissolved gas analysis integrated with thermodynamics theory', *IET Sci. Meas. Technol.*, 2018, **12**, (3), pp. 388–394

[28] Rogers, R.R.: 'IEEE and IEC codes to interpret incipient faults in transformers, using gas-in-oil analysis', *IEEE Trans. Electr. Insul.*, 1978, **EI-13**, (5), pp. 349–354

[29] Ibrahim, S., Taha, I.B.M., Ghoneim, S.M.: 'DGA tool GitHub repository'. https://github.com/Saleh860/DGA, accessed August 2017

10 Appendix

10.1 Duval triangle method

Michel Duval developed this method in 1974. He used the test samples that results from the faulted transformer (1000 samples) and constructing his triangle based on the dissolved gas concentrations associated with each fault type. The triangle is built based on the percentage of three gases (CH_4, C_2H_4, and C_2H_2) with the sum of these gases to specify the transformer fault. Fig. 1 explained the zone limits of each type.

© The Institution of Engineering and Technology 2018

17518695, 2018, 18, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-gtd.2018.5564 by Capes, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.2 Rogers' 4-ratio method

The proposed ratios are based on five combustible gases (H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2). The ratios are ($R_1 = CH_4/H_2$, $R_2 = C_2H_2/C_2H_4$, $R_3 = C_2H_4/C_2H_6$, and $R_4 = C_2H_6/CH_4$) and the coded values of these ratios are specified based on the ratio limits as in [6]. The fault types are also determined when the code is achieved [6].

10.3 IEC 60599 code

The IEC code [1] is based on three ratios as ($R_1 = CH_4/H_2$, $R_2 = C_2H_2/C_2H_4$, $R_3 = C_2H_4/C_2H_6$). The coded value and ratio limits are explained in [1]. The faults types correspond to the coded value are demonstrated in [1].

10.4 Clustering method

The clustering technique [14] is used to diagnose the transformer fault based on not only the concentration of the transformer oil-dissolved gases with key gas for each fault type but also newly proposed gas ratios to separate the interference that occurs among different fault types. The main objective of the technique is overcoming the undetermined fault types in addition to the interference among the fault types that occurs with the traditional methods. The technique is built based on the main five combustible gases (H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2). The proposed ratios are explained as follows [14]:

$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \\ r_7 \end{pmatrix} = \begin{pmatrix} C_2H_2/H_2 \\ C_2H_2/CH_4 \\ C_2H_2/C_2H_6 \\ C_2H_4/H_2 \\ C_2H_4/CH_4 \\ (C_2H_4/H_2) + (C_2H_4/CH_4) \\ C_2H_4/C_2H_6 \end{pmatrix} \quad (1)$$

The proposed technique is based on some limits and ratios limit of the gas concentrations as in [14].

10.5 CSUS ANN method

The ANN of this method is built based on the measured dissolved gas concentration for 386 samples with specified fault types. The used dissolved gases are as in (2) [6]

$$[X_{GC}] = [H_2, CH_4, C_2H_6, C_2H_4, C_2H_2]^T \quad (2)$$

The ANN of the CSUS are based on 309 test case samples and tested with 76 test samples.

10.6 Conditional probability method

The conditional probability method [15] is built based on five combustible gases (H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2). The percentage of these gases according to their summation is computed to specify the transformer fault types. The fault types are classified as PD, low-energy discharge, high-energy discharge, low thermal, medium thermal, and high thermal. The occurrence and non-occurrence probabilities of the fault types based on the percentage of gas consternations are computed. Therefore, the occurrence probability of each fault type is estimated. The method proposed three scenarios to specify the fault type, and according to the results of 403 samples the best scenario is selected [15]. The three scenarios are mentioned in [15], which illustrates the different stages required for each scenario. In this software package, the scenario (2) mentioned in [15] are used to detect the oil transformer fault type.

10.7 Modified Rogers' ratios and IEC code

Rogers' 4-ratio and IEC 60599 code methods have some limitations for interpreting the transformer oil faults. Therefore, modified Rogers' 4-ratio and IEC 60599 codes are presented in [26]. The modified codes are proposed based on 386 actual transformer oil test samples that are collected from the electric utility in Egypt and literature. The samples are categorised to six fault types; PD, low-energy discharge (D1), high-energy discharge (D2), low thermal fault (T1), medium thermal fault (T2), and finally high thermal fault (T3). The modified code flowchart of Rogers' 4-ratio and IEC 60599 codes are depicted in [26].

© The Institution of Engineering and Technology 2018

17518695, 2018, 18, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/iet-gtd.2018.5564 by Capes, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License